Roll No.....

DRONACHARYA COLLEGE OF ENGINEERING, GURUGRAM
PRE-UNIVERSITY EXAMINATION APRIL, 2025
(Gurugram University)

Semester: IV (B Tech) **Branch: CSE-IOT**

QUESTION PAPER

Subject with Code: Programming in Java (21135)

Time: 3 hours **Max. Marks: 70**

Note: *Q.1 is compulsory. Attempt any ONE question from each Unit. Each question carries of 14 marks. Assume data, if any missing. Mention CO (COURSE OUTCOME) against each question.*

Q. 1. Objective type / Short-answer type questions. **(2x7=14)**

i. Define Java Virtual Machine. How does it ensure platform independence? (CO1)

ii. Differentiate between method overloading and method overriding. (CO2)

iii. What is the role of the this keyword in Java? (CO1)

iv. List any two differences between Runnable interface and Thread class. (CO5)

v. What is the use of try-catch block in Java? Give an example. (CO4)

vi. What are generics in Java? Mention one use-case. (CO6)

vii. Differentiate between Enumerator and Iterator. (CO4)

UNIT-I

Q. 2 a) Explain the compilation and execution process of a Java program with a diagram. (CO1)

b) Describe JVM, JDK, and JRE. How are they interrelated? (CO1)

Q.3 a) Discuss different data types used in Java with the help of suitable example. (CO1)

b) Explain Java security architecture with the help of the sandbox model. (CO1)

UNIT-II

Q. 4 a) What is inheritance? How does Java support code reusability through inheritance? Illustrate with an example. (CO2)

b) Explain method overloading and method overriding with proper examples. (CO2)

Q.5 a) Differentiate between abstract classes and Interface. Explain with the help of Example Code. (CO2)

b) What are anonymous inner classes? How are they useful in Java applications? (CO1)

UNIT-III

Q. 6 What is thread synchronization? How can Producer Consumer problem be implemented in Java? (CO5)

Q.7 a) Compare Swing and AWT. What are the advantages of using Swing in GUI development? (CO6)

b) WAP to create a Login Page using Java Swing and proper login verification code to verify login details must be implemented. (CO5)

UNIT-IV

Q. 8 a) What is Java Collection Framework? Discuss the role and importance of this framework in designing different applications. (CO6)

b) What is the difference between Comparable and Comparator? Illustrate with code. (CO6)

Q.9 a) Discuss different implementations of List and Set in Java Collection Framework (CO6)

b) Write Java code to implement an ArrayList and a Hashset of String Data values. Also write code to add elements and iterate elements of these data structures. (CO6)